Assignment-2

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Answer to the question no : 1

A subset (W) of a vector space $\mathbb{R}^3$ is a subspace of it satisfies the following three conditions:

i) The zero vector of $(\mathbb{R}^3)$ is in (W).
ii) (W) is closed under vector addition.
iii) (W) is closed under scalar multiplication.

(i) The zero vector in $\mathbb{R}^3$ is $(0,0,0)$

For $(1, x_2, x_3)$ to include the zero vector, there must be values (x_2) and (x_3) such that, $((1, x_2, x_3) = (0,0,0))$.

Let, $0 = (0,0,0) \in W$
$1+0+0 = 1 \neq 0$

However, there are no (x_2) and (x_3) such that $(1=0)$.

Hence the zero vector is not in (W)

$\therefore$ W is not a subspace of $(\mathbb{R}^3)$

(ii) Let, $u = ((1, u_2, u_3) \in W)$ and $v = ((1, v_2, v_3) \in W)$.

Then, $(u+v) = (1+1, u_2+v_2, u_3+v_3) = (2, u_2+v_2, u_3+v_3)$

Since, $(2, u_2+v_2, u_3+v_3)$ is not the form $(1, x_2, x_3)$.

(W) is not closed under vector addition.

(iii) Let, $u = ((1, u_2, u_3) \in W)$ and $(c \in \mathbb{R})$

Then, $(cu = c(1, u_2, u_3) = (c, cu_2, cu_3))$

Since, (c, cu_2, cu_3) is not the form $(1, x_2, x_3)$ unless $(c=1)$, (W) is not under scalar multiplication.

$\therefore$ (W) is not a subspace of $\mathbb{R}^3$.

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Answer to the question no: 2

(i)

Let,

$x, y \in V$

$x+y \in V$

Given, $Ax = 5x$ and

let, $Ay = 5y$

Now, we have, $A(x+y) = Ax + Ay$

$= 5x + 5y = 5(x+y)$

Thus, $x+y \in V$

Let,

c be any scalar.

$Ax = 5x$

$\therefore A(cx) = cAx = c(5x) = 5(cx).$

Thus, $cx \in V$

Here,

V containing the zero vector

because $A(0) = 0 = 5(0)$

$\therefore$ V is a subspace of R^2

(ii)

Let, $x = \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}$

$\therefore Ax = 5x$

$$\Rightarrow \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = 5 \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4a + b \\ 3a + 2b \end{bmatrix} = \begin{bmatrix} 5a \\ 5b \end{bmatrix}$$

$\therefore 4a + b = 5a$

$3a + 2b = 5b$

$\therefore b = 5a - 4a$

$\Rightarrow a = b$

$\therefore 3a = 3b$

$\Rightarrow a = b$

$$\therefore x = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a \\ a \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\therefore$ basis for V is $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ and dimension of V is 1.

$\therefore \dim(V) = 1$

Answer to the question no 3

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix} \quad r_1 \longleftrightarrow r_2$$

$$= \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix} \quad \begin{matrix} r_3' = r_3 - 3r_1 \\ r_4' = r_4 - r_1 \end{matrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix} \quad \begin{matrix} r_3' = r_3 - r_2 \\ r_4' = r_4 - r_2 \end{matrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ Basis of $R(A)$ is $\{(1,0,1,1), (0,1,-3,-1)\}$

$\therefore$ Basis of $C(A)$ is $\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}\right\}$

$\therefore \dim(R(A)) = 2$

$\dim(C(A)) = 2$

$\therefore$ rank $= 2$

(Ans)

Answer to the question no 4

$$A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix} \begin{array}{l} r_2' = r_2 - 2r_1 \\ r_3' = r_3 + r_1 \end{array}$$

~~$N(A) = \{ x \in \mathbb{R}^4$~~

$$= \begin{bmatrix} 1 & 4 & 5 & 2 \\ 0 & -7 & -7 & -4 \\ 0 & 7 & 7 & 4 \end{bmatrix} \begin{array}{l} r_2' = \frac{r_2}{-7} \\ r_3' = \frac{r_3}{7} \end{array}$$

$$= \begin{bmatrix} 1 & 4 & 5 & 2 \\ 0 & 1 & 1 & \frac{4}{7} \\ 0 & 1 & 1 & \frac{4}{7} \end{bmatrix} r_3' = r_3 - r_2$$

$$= \begin{bmatrix} 1 & 4 & 5 & 2 \\ 0 & 1 & 1 & \frac{4}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore x_1 + 4x_2 + 5x_3 + 2x_4 = 0$$

$$x_2 + x_3 + \frac{4}{7}x_4 = 0$$

free variable
$\hookrightarrow x_3, x_4$

Let,

$x_3 = s$

$x_4 = t$

$$\therefore x_2 + s + \frac{4t}{7} = 0$$

$$\Rightarrow x_2 = \left(-s - \frac{4t}{7}\right)$$

$$\therefore x_1 + 4\left(-s - \frac{4t}{7}\right) + 5s + 2t = 0$$

$$\Rightarrow x_1 - 4s - \frac{16t}{7} + 5s + 2t = 0$$

$\Rightarrow x_1 + s - \frac{2}{7}t = 0$

$\Rightarrow x_1 = -s + \frac{2}{7}t$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -s + \frac{2}{7}t \\ -s - \frac{4}{7}t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{pmatrix}$$

So, the null space of A is,

$$N(A) = \left\{ s \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{pmatrix} \right\}$$

basis of null space of A,

$$N(A) = \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{pmatrix} \right\}$$

Ans

Answer to the question no 5

The vector space V is defined as $\mathbb{R}^2$ with the usual addition of vectors and scalar multiplication. However, the multiplication of cu is defined to produce $(cu_1, 0)$ instead of (cu_1, cu_2)

Conditions:

(i) $u+v \in V$ for all $u, v \in V$

$$u+v = (u_1 + v_1, u_2 + v_2)$$

This is satisfied because the addition is defined as usual addition in $\mathbb{R}^2$

(ii) $u+v = v+u$:

$$(u_1+v_1, u_2+v_2) = (v_1+u_1, v_2+u_2)$$

$= v+u$, so addition is commutative.

This is satisfied because the addition is defined as usual addition in $\mathbb{R}^2$

(iii) $u+(v+w) = (u+v)+w$:

$u+(v+w) = (u_1+(v_1+w_1), u_2+(v_2+w_2))$ and

$(u+v)+w = ((u_1+v_1)+w_1, (u_2+v_2)+w_2)$

so, these are equal.

This is satisfied because the addition is defined as usual addition in $\mathbb{R}^2$

(iv) There exist a $0 \in V$ such that $u+0=u$ for all $u \in V$

There exist $0 \in \mathbb{R}^2$ such that,

$u+0 = (u_1+0, u_2+0) = (u_1, u_2)$ for all $u \in \mathbb{R}^2$.

so,

This is satisfied because the zero vector is $(0,0)$.

(v) There exists a $-u \in V$ for all $u \in V$, such that $u+(-u)=0$.

There exist $-u \in \mathbb{R}^2$ such that $u+(-u)$
$= (u_1+(-u_1), u_2+0) = (0,0)$ for all $u \in \mathbb{R}^2$

This is satisfied.

(vi) $ku \in V$ for all $u \in V$ and $k \in F$

$k.u = (k.u_1, 0)$.

So, this is satisfied because, scalar multiplication is defined as $(ku_1, 0)$

(vii) $a(u+v) = au + av$ for all $a \in F$ and $u, v \in V$:

$a(u+v) = (a(u_1+v_1, u_2+v_2) = (a(u_1+v_1), 0)$,

and, $au + av = (au_1, 0) + (av_1, 0)$

$= (au_1 + av_1, 0)$

These are equal, so, it satisfies the distributive property.

(viii) $(a+b)u = au + bu$ for all $a, b \in F$ and $u \in V$:

$(a+b)u = ((a+b)u_1, 0)$ and $au + bu = (au_1, 0) + (bu_1, 0)$

$= ((a+b)u_1, 0)$

These are equal, so it satisfies the distributive property.

also, this is satisfied because scalar multiplication is defined element-wise.

(ix) $(ab)u = a(bu)$ for all $a, b \in F$ and $u \in V$:

$(ab)u = ((ab)u_1, 0)$ and $a(bu) = (a(bu_1), 0)$

This is not satisfied. The multiplication of cu produces $(cu_1, 0)$, while $(ab)u$ would produce (abu_1, abu_2), and these are not equal.

(x) $1u = u$, where $1 \in F$ and for all $u \in V$:

This is satisfied because $1u = (1u_1, 0)$
$= (u_1, 0) = u$

Therefore,

condition (ix) is one that is not satisfied in the modified vector space.

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